



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Merge Manor

CSA0920-PROGRAMMING IN JAVA

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Review 1

01/08/2026

GAME CONCEPT OVERVIEW



What is Merge Manor?

- A puzzle game where players merge two identical items.
- Two same-level items combine to create a higher-level item.
- The player keeps upgrading items to complete goals.
- Core gameplay loop: Collect → Merge → Upgrade → Complete Goal
→ Repeat
- Win by creating the required target item.



REFERENCE GAME STUDY



Reference Game - Merge Mansion

- Popular merge puzzle game.
- Simple drag-and-drop gameplay.
- Progress through item upgrades.
- Goal-based missions increase engagement.

Our Simplified Version

- Fixed item hierarchy.
- Basic merge rules.
- No animations.
- Simple drag-and-drop interface.



OBJECTIVES / LEARNING OUTCOMES



- Understand Object-Oriented Programming.
- Use Java Collections Framework.
- Implement drag-and-drop GUI.
- Apply Exception Handling.
- Store game progress using JDBC.



PROPOSED MODULES



Module 1:

Item Hierarchy & Merge Rule Engine

- Create different item levels.
- Merge two identical items into the next-level item.
- Validate whether merging is allowed.

Java Concept:

OOP (Inheritance, Classes, Objects)



PROPOSED MODULES



Module 2:

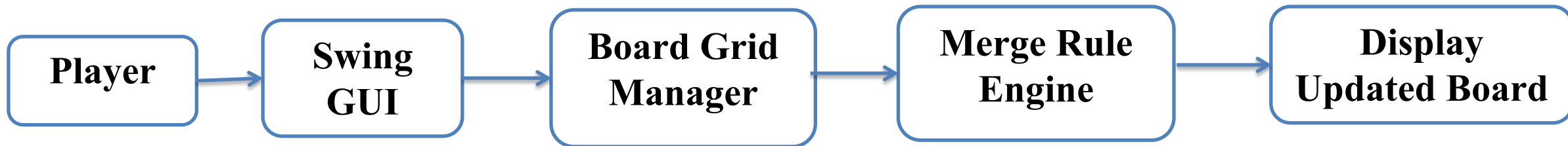
Board Grid Manager (Collections)

- Store all game items on the board.
- Add, remove and update items after merging.
- Maintain the board state using Java Collections.

Java Concept:

ArrayList / Collections Framework

System Architecture



TECH STACK & WHY

Technology	Purpose
Java	Game development
OOP	Item hierarchy
Collections	Store board items
Swing	User Interface
Exception Handling	Handle invalid merges
JDBC	Save progress (future module)

PROPOSED DATABASE SCHEMA

Table: PLAYER_PROGRESS

Column	Data Type
Player_ID	INT
Player_Name	VARCHAR(50)
Board_State	TEX
Last_Played	DATE

Stored Data

- Current board layout
- Player information
- Last saved progress



MULTITHREADING PLAN



- Keep the GUI responsive.
- Refresh the game board after each merge.
- Handle future animations without freezing the interface.

EXCEPTION HANDLING PLAN

Situation	Exception	Solution
Invalid merge	IllegalArgumentException	Show error message
Empty board cell	NullPointerException	Validate before merging
Invalid user action	Exception	Prevent invalid moves



WORK PLAN



Week 1

- Design Item Hierarchy
- Implement Merge Rules

Week 2

- Develop Board Grid Manager
- Test merge operations

Week 3

- Build Swing GUI

Week 4

- JDBC integration
- Testing and debugging



ANTICIPATED CHALLENGES



Challenge 1

- Implementing correct merge rules.

Solution: Use conditional logic to validate merges.

Challenge 2

- Managing the board efficiently.

Solution: Use Java Collections to update the board dynamically.

Challenge 3

- Keeping the GUI responsive.

Solution: Use proper event handling and multithreading where required.

Thank You!